UNIT 1 · LESSON 4 · INPUT

# Binary Numbers

Date: _____

○ = 0 ☐ = 1 Last lesson's circles and squares **were binary**. We just swap the shapes for digits.

## Vocabulary

**bit**

**byte**

**binary number**

**place value**

## Place values: each spot is worth more than the one to its right

| Decimal | 1000 | 100 | 10 | 1 |
|---|---|---|---|---|
| Multiply by | ×10 | ×10 | ×10 | start |
| **Binary** | 8 | 4 | 2 | 1 |
| Multiply by | ×___ | ×___ | ×___ | start |

### Binary → decimal

Add up the place values that have a 1.

| 8 | 4 | 2 | 1 |
|---|---|---|---|
| 1 | 0 | 1 | 1 |

8 + 0 + 2 + 1 = **11**

### Decimal → binary

Write the number above the biggest place value. Is the place value ≤ it? Yes → write 1, subtract, carry what's left right. No → write 0, carry the same number.

| 25 | 9 | 1 | 1 | 1 |
|---|---|---|---|---|
| **1** | **1** | **0** | **0** | **1** |
| 16 | 8 | 4 | 2 | 1 |

$25_{dec}$ = $\mathbf{11001}_{bin}$

**KEY IDEA** Binary works just like decimal, but each place value is worth **double** the one to its right instead of ten times.